

1. Let $A = \{1, 2, 3, 5\}$ and $B = \{a, b, 2, c, 5\}$
 - a. Find $A \cap B$
 $= \{2, 5\}$
 - b. Find $A \cup B$.
 $= \{1, 2, 3, 5, a, b, c\}$
 - c. Find $B - A$.
 $= \{a, b, c\}$
 - d. Find $A - B$.
 $= \{1, 3\}$
2. Let A be a nonempty subset of B . Which of the following is true?

~~(a)~~ $A \cap B = B$.

~~(b)~~ $A \cup B = A$.

(c) $A \cup B = B$.

~~(d)~~ $A \cap B = \emptyset$.
3. Let $A = \{3, 5\}$ and $B = \{a, b, 5\}$.
 - a. List all possible subsets of B .
 $\{\emptyset\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, 5\}, \{b, 5\}, \{a, b, 5\}$
 - b. Find $A \times B$.
 $\{(3, a), (3, b), (3, 5), (5, a), (5, b), (5, 5)\}$
4. Prove that $A \cup \emptyset = A$.
Proof. $\emptyset \subseteq A$ and $A \subseteq A$ thus $A \cup \emptyset \subseteq A$ and $A \subseteq A \cup \emptyset$.
 Alternatively,
 $(A \subseteq A \cup \emptyset)$ Let $a \in A$ then $a \in A$ or $\emptyset$ thus $a \in A \cup \emptyset$.
 $(A \cup \emptyset \subseteq A)$ If $A = \emptyset$ then $A \cup \emptyset = \emptyset \subseteq A$. Otherwise if $a \in A \cup \{\emptyset\}$ then $a \in A$. Thus $A \cup \{\emptyset\} = A$. □
5. Consider the map $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ where $f(x, y) = xy$. Prove that f is onto.
Proof. Let $z \in \mathbb{R}$. $1 \in \mathbb{R}$ so then $f(1, z) = 1 \cdot z = z$. Thus f is onto. □
6. Define the relation $\mathcal{R}$ on the real number set $\mathbb{R}$ as $x\mathcal{R}y \iff xy \geq 0$.
 - a. Show that $\mathcal{R}$ is reflexive and symmetric.
Proof. Reflexive: $x \cdot x = x^2 \geq 0$.
 Symmetric: If $x \cdot y \geq 0$ then $x \cdot y = y \cdot x \geq 0$. □
 - b. Is $\mathcal{R}$ an equivalence relation? Explain.
 No. $\mathcal{R}$ is not transitive. $-1 \cdot 0 \geq 0$ and $0 \cdot 1 \geq 0$ but $-1 \cdot 1 = -1 < 0$.
7. Let X be a nonempty set. Define the following relation on the set of subsets of X as follows $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$.
 - a. Show that $\mathcal{R}$ is symmetric.
Proof. Suppose that $A \cap B \neq \emptyset$ then $B \cap A \neq \emptyset$. □
 - b. Show that $\mathcal{R}$ is not reflexive.
Proof. $\emptyset \in X$, and $\emptyset \cap \emptyset = \emptyset$. □
8. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions.
 - a. If $g \circ f$ is one-to-one then

~~(a)~~ f is onto.

(b) f is one-to-one.

(c) g is one-to-one.

(d) Both f and g are one-to-one.

~~(e)~~ None of the above.
 - b. If $|A| > |B|$ and $A \neq \emptyset$ can f be one-to-one?
 No. By the Pigeon Hole Theorem.

9. Let S_3 be the symmetric group on three elements.

a. Which of the following is true?

- (a) S_3 has at least one non-abelian subgroup.
~~(b)~~ All proper subgroups of S_3 are abelian.
~~(c)~~ S_3 is abelian.
~~(d)~~ S_3 is cyclic.

b. What is the order of S_3 and S_4 ?

$$|S_3| = 3! = 6$$

$$|S_4| = 4! = 24$$

10. Let $f : G_1 \rightarrow G_2$ be a group homomorphism and let $H \leq G_2$. Recall that $f^{-1}(H) = \{g \in G_1 | f(g) \in H\}$.

a. Which of the following is true?

- ~~(a)~~ $f^{-1}(H)$ consists only of the identity of G_1 .
 (b) $f^{-1}(H)$ is a subgroup of G_1 .
~~(c)~~ $gf^{-1}(H) = f^{-1}(H)g$ for all $g \in G_1$ (the left and right cosets are equal).
~~(d)~~ $f^{-1}(H)$ is not a subgroup of G_1 .

b. Let $a, b \in G_1$. Does $f(ab) = f(a)f(b)$?

Yes. Since f is homomorphic $f(ab) = f(a)f(b)$ by definition.

11. Let G be a finite group of order 12 and $g \in G$.

a. The order of g cannot be:

- (a) 8
~~(b)~~ 6
~~(c)~~ 4
~~(d)~~ 3

b. Let $|H| = 6$. Does Lagrange's theorem imply that $H \leq G$?

No. Lagrange's Theorem is a necessary but not sufficient condition for a subset to be a subgroup. For example, the alternating group A_4 has 12 elements but no subgroup of order 6.

12. Let G be a group of order 4. Which of the following is true?

- ~~(a)~~ G is a cyclic group.
~~(b)~~ G is $\mathbb{Z}_2 \times \mathbb{Z}_2$.
~~(c)~~ G is $\mathbb{Z}_4$.
 (d) G is abelian.

13. Let $G = \{e, a_1, a_2, a_3, a_4, a_5\}$ be a group where e is the identity.

a. Find a subgroup of G .

The trivial group e is a subgroup of every group G .

b. Find a subgroup of G distinct from the group given for part (a).

$\langle a_i \rangle \leq G$, and $\langle a_i \rangle \neq e$ for any $1 \leq i \leq 5$.

c. Can $|a_2| = 5$?

No. By Lagrange's theorem $|a_2| \mid |G|$ and $5 \nmid 6$

14. Let H and K be subgroups be non-trivial subgroups of G . Prove or disprove that $H \cap K$ is a subgroup of G .

Proof. $H \cap K \leq G$. Let $a, b \in H \cap K$. Then $b^{-1} \in H$ and K since $b \in H$ and K and $H, K \leq G$. Thus $ab^{-1} \in H \cap K$ because $a \in H$ and K and subgroups are closed under their operation. $\square$

15. Let R be a ring and $a, b \in R$.

a. Which statement is true?

- ~~(a)~~ If $ab = 0$ then $a = 0$ or $b = 0$.
~~(b)~~ If $a^2 = 0$ then $a = 0$.
~~(c)~~ If $a \neq 0$ and $b \neq 0$ then $ab \neq 0$.
 (d) None of the above.

b. Show that $\mathbb{Q}$, the set of rational numbers, is a ring under addition and multiplication.

Proof. $\mathbb{Q}$ is an abelian group. $\mathbb{Q} \subset \mathbb{R}$ so then associativity and distributivity hold under multiplication. Thus $\mathbb{Q}$ is a ring. $\square$